LuMay AI

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ML to Agentic AI Roadmap







Reshmawithai

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Cover

ML to Agentic AI Roadmap.

From basic ML models to AI agents that think, plan, use tools, and take action.

```python
roadmap = [
    "ML Foundations",
    "Deep Learning",
    "NLP + Transformers",
    "LLMs",
    "RAG",
    "Agentic AI"
]
```



Step 1

Start with ML Foundations

Before agents, master the basics:

1. Supervised & Unsupervised Learning
2. Feature Engineering
3. Model Evaluation
4. Linear Models, Trees, Boosting

Python

Run

```python
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

X_train, X_test, y_train, y_test = train_test_split(X, y)

model = RandomForestClassifier()
model.fit(X_train, y_train)

print(model.score(X_test, y_test))
```



Step 2

Learn Deep Learning

Understand how neural networks learn:

1. Neural Networks
2. Backpropagation
3. CNNs, RNNs, LSTMs
4. PyTorch / TensorFlow

Python

Run

```python
import torch
import torch.nn as nn

model = nn.Sequential(
    nn.Linear(10, 32),
    nn.ReLU(),
    nn.Linear(32, 1)
)

output = model(torch.randn(1, 10))
```



Step 3

Enter NLP & Transformers

This is where modern AI begins:

1. Text preprocessing
2. Embeddings
3. Transformer architecture
4. Hugging Face ecosystem

```python
from transformers import pipeline

classifier = pipeline("sentiment-analysis")

result = classifier("Agentic AI will change workflow.")
print(result)
```



Step 4

Master LLMs

Move to Generative AI by learning:

1. Tokens & Attention
2. Prompt Engineering
3. Fine-tuning
4. Open vs Closed Models

```python
name = "AI"
# AI is the future.
"""Learn AI to change worlds with our example.
Stay curious."""

numbers = [1, 2, 3, 4]  # Example list
print(numbers)
```

Step 5

Build with RAG Systems

Learn the memory + knowledge layer:

1. Vector Databases
2. Retrieval Pipelines
3. Chunking + Embeddings
4. LlamaIndex / LangChain

```python
query = "What is our refund policy?"

docs = vector_db.search(query, top_k=3)

answer = llm.generate(
    context=docs,
    question=query
)
```



Step 6

Step into Agentic AI

Now combine reasoning + action:

1. Tool usage
2. Planning
3. Multi-step reasoning
4. CrewAI, LangGraph, AutoGen

```python
def _score(
    pred: "AnswerLike | Result",
    gold: list[dict], summary:List]
)

result = _score.run(
    "Find 5 AI trends for 2025"
)
```

Step 7

Learn Agent Workflows

Agents are not just chatbots. They execute workflows.

1. Understand the goal
2. Break it into steps
3. Use tools
4. Return final output

```python
task = "Create content on AI agents"

steps = planner.break_down(task)

for step in steps:
    result = agent.execute(step)

final_output = agent.summarize(results)
```

Step 8

Add Memory to Agents

Good agents remember context.

1. Short-term memory
2. Long-term memory
3. User preferences
4. Conversation history

```python
memory.save({
    "user_goal": "Learn Agentic AI",
    "level": "Beginner",
    "preferred_style": "Simple examples"
})

context = memory.load("user_goal")
```



Step 9

Connect Agents to Tools

Real agents interact with systems.

1. Search tools
2. Database tools
3. CRM / ERP tools
4. Email, Calendar, APIs

```python
def get_customer_data(customer_id):
    return crm_search(customer_id)

query_tool_tool = {
    name: "CRM Search",
    func: get_customer_data
}

agent.run("Find customers under lifetime > 3")
```

Step 10

Build Multi-Agent Systems

One agent is useful. A team of agents is powerful.

1. Research Agent
2. Planner Agent
3. Writer Agent
4. Reviewer Agent

```python
research = agent("Researcher")
writer = agent("Content Writer")
reviewer = agent("Editor")

data = research.run(topic)
draft = writer.run(data)
final = reviewer.run(draft)
```

Step 11

Learn AI Evaluation

Agent quality depends on testing.

1. Accuracy testing
2. Hallucination checks
3. Tool-call success rate
4. Human feedback loops

```python
question = "What are retrieval + generation"
actual_tool = get_tool("Ask_Dr_RAG")

score = evaluate_response(
    expected_results,
    actual_status
)
print(score)
```

Step 12

Deploy Real AI Agents

Move from demo to production.

1. Monitoring
2. Logs
3. Guardrails
4. Cost optimization

```python
@app.post("/agent")
def run_agent(request):
    response = agent.run(request.query)

    logger.info(response)
    return {"answer": response}
```

Your Roadmap is Simple

ML → Deep Learning → NLP → LLMs → RAG → Agents

Start with fundamentals.
Build small projects.
Connect tools.
Deploy real workflows.



The future belongs to people who can build AI that acts.

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